Kepler’s laws

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use orbital shape, equal areas and period comparisons.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 11.*
3 s to top; 6 s total, assuming no drag and equal launch/landing heights.

02. Worked example: explain the model
Faster near perihelion, slower near aphelion; Sun at focus.

03. Read and interpret the model
Fastest closest; slowest farthest. Sun at focus, not center.

04. Apply the physics
Outer; 16 years.

05. Explain your reasoning
No; need orbital phase, time reference, geometry, and Earth’s position, with uncertainty.

06. Review: connect an earlier idea *Review: Unit 2, Day 2.*
50 m/s downward.

07. Exit check
A; equal areas in equal times.

08. STRETCH: test the model
Period grows faster than a: $T^2=a^3$, so the given size factor 4 produces period factor 8.

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01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 11.*
Both ranges≈68.94 m. Times≈2.536 s and 5.438 s. The larger angle gives longer flight.

02. Worked example: explain the model
Faster near perihelion, slower near aphelion; Sun at focus.

03. Read and interpret the model
Fastest closest; slowest farthest. Sun at focus, not center.

04. Apply the physics
$T=\sqrt{1.42^3}\approx1.69$ years; 8 years at 4 AU; T∝a^1.5.

05. Explain your reasoning
No; need orbital phase, time reference, geometry, and Earth’s position, with uncertainty.

06. Review: connect an earlier idea *Review: Unit 2, Day 2.*
v_f=34 m/s down; Δy=57 m down.

07. Exit check
2^1.5≈2.83; central mass/unit normalization differs.

08. STRETCH: test the model
a=9 AU; perihelion 4.5 AU, aphelion 13.5 AU. Formula is supplied for the extension.